Score Driven Models - Lista 2

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Questão 1

$$p(y_t|y_{t-1}) \sim NB(\mu, r)$$

Item a) Parametrização e momentos da distribuição

1a) $p(y_t|y_{t-1}) \sim NB(\mu, r)$, $r > 0$ e $\mu > 0$

• $E(y_t|y_{t-1}) = \mu_{t|t-1}$ //

$$\mu = \frac{r(1-p)}{p} \Rightarrow \mu p + r p = r \Rightarrow p = \frac{r}{\mu + r}, \quad 1 - p = \frac{\mu}{\mu + r}$$

• $Var(y_t|y_{t-1}) = \dfrac{\mu_t}{p} = \dfrac{\mu_t(\mu_t + r)}{r}$

• $S(y_t|y_{t-1}) = \dfrac{2-p}{\sqrt{(1-p)r}} = \left(2 - \dfrac{r}{\mu_t + r}\right)\sqrt{\dfrac{(\mu_t + r)}{\mu_t r}}$

$$\Rightarrow S(y_t|y_{t-1}) = \frac{2\mu_t + r}{\sqrt{(\mu_t + r)\mu_t r}}$$

• $K(y_t|y_{t-1}) = 3 + \dfrac{6}{r} + \dfrac{p^2}{(1-p)r} = 3 + \dfrac{6}{r} + \dfrac{r^2}{(\mu + r)^2} \cdot \dfrac{\cancel{(\mu_t + r)}}{\mu_t r}$

$$= 3 + \frac{6\mu_t(\mu_t + r) + r^2}{(\mu_t + r)\mu_t r}$$

$$\Rightarrow p(y_t|y_{t-1}) = \binom{y_t + r - 1}{y_t}\left(\frac{\mu_t}{\mu_t + r}\right)^{y_t}\left(\frac{r}{\mu_t + r}\right)^{r}$$

Item b) Equação da média com GAS(p,q)

i) Link identidade

b) i) (1) link identidade

$$\mu_{t+1|t} = \omega + \sum_{i=1}^{p} \alpha_i \mu_{t-i+1|t-i} + \sum_{j=1}^{q} \beta_j s_{t-j+1}, \quad s_t = (\mathcal{I}_{t|t-1})^{-d} \nabla_t$$

1b) i) continuação

$$L(\mu) = \binom{y+r-1}{y} \left(\frac{r}{r+\mu}\right)^{r} \left(\frac{\mu}{r+\mu}\right)^{y}$$

$$\Rightarrow l(\mu) = \ln\binom{y+r-1}{y} + r\ln r + y\ln\mu - (r+y)\ln(r+\mu)$$

$$\nabla_t = \frac{\partial l(\mu)}{\partial \mu} = \frac{y_t}{\mu_t} - \frac{r+y_t}{r+\mu_t}$$

$$\frac{\partial \nabla_t}{\partial \mu} = -\frac{y_t}{\mu_t^2} + \frac{r+y_t}{(r+\mu_t)^2}$$

$$\Rightarrow \mathcal{I}_{t|t-1} = -E_{t-1}\left(\frac{-y_t}{\mu_t^2} + \frac{r+y_t}{(r+\mu_t)^2}\right) = \frac{E(y_t|y_{t-1})}{\mu_{t|t-1}^2} - \frac{r+E(y_t|y_{t-1})}{(r+\mu_{t|t-1})^2}$$

$$\Rightarrow \mathcal{I}_{t|t-1} = \frac{1}{\mu_{t|t-1}} - \frac{\cancel{r+\mu_{t|t-1}}}{(r+\mu_{t|t-1})^2} = \frac{1}{\mu_{t|t-1}} - \frac{1}{r+\mu_{t|t-1}}$$

$$\Rightarrow \mathcal{I}_{t|t-1} = \frac{r}{\mu_{t|t-1}(r+\mu_{t|t-1})}$$

para $d = 1$,

$$s_t = \frac{\mu_t(r+\mu_t)}{r}\left(\frac{y_t}{\mu_t} - \frac{r+y_t}{r+\mu_t}\right) = \frac{(r+\mu_t)y_t}{r} - \frac{\mu_t(r+y_t)}{r}$$

$$\Rightarrow s_t = y_t - \mu_{t|t-1}$$

$$\Rightarrow \mu_{t+1|t} = \omega + \sum_{i=1}^{p} \alpha_i \mu_{t-i+1|t-i} + \sum_{j=1}^{q} \beta_j (y_{t-j+1} - \mu_{t-j+1|t-j})$$

ii) Link log

1b) i) (2) link log

$$\log(\mu_{t|t-1}) = \tilde{\mu}_{t|t-1} \Rightarrow \mu_t = e^{\tilde{\mu}_t}$$

$$\tilde{\mu}_{t+1|t} = \omega + \sum_{i=1}^{p} \alpha_i \tilde{\mu}_{t+1-i|t-i} + \sum_{j=1}^{q} \beta_j \tilde{s}_{t-j+1}$$

$$\tilde{\nabla}_t = (\dot{h}_t)^{-1} \nabla_t$$

$$= \left(\frac{\partial \log(\mu_{t|t-1})}{\partial \mu} \right)^{-1} \nabla_t = \left(\frac{1}{\mu_{t|t-1}} \right)^{-1} \nabla_t$$

$$\Rightarrow \tilde{\nabla}_t = \mu_{t|t-1} \nabla_t$$

$$\tilde{I}_{t|t-1} = (\dot{h}_t)^{-1} I_{t|t-1} (\dot{h}_t)^{-1}$$

$$= \mu^2_{t|t-1} I_{t|t-1}$$

p/ $d = 1$, $\tilde{s}_t = (\tilde{I}_{t|t-1})^{-1} \tilde{\nabla}_t$

$$\tilde{s}_t = \frac{1}{\cancel{\mu_t^2}} \frac{\cancel{\mu_t}(r+\mu_t)}{r} \cdot \cancel{\mu_t} \left(\frac{y_t}{\mu_t} - \frac{r+y_t}{r+\mu_t} \right)$$

$$= \frac{(r+\mu_t) y_t}{r \mu_t} - \frac{r+y_t}{r}$$

$$= \frac{\cancel{r}(y_t - \mu_t)}{\cancel{r} \mu_t}$$

$$\Rightarrow \tilde{s}_t = y_t e^{-\tilde{\mu}_t} - 1$$

$$\Rightarrow \tilde{\mu}_{t+1|t} = \omega + \sum_{i=1}^{p} \alpha_i \tilde{\mu}_{t+1-i|t-i} + \sum_{j=1}^{q} \beta_j \left(y_{t-j+1} e^{-\tilde{\mu}_{t+1-j|t-j}} - 1 \right)$$

$$\Rightarrow \ln \mu_{t+1|t} = \omega + \sum_{i=1}^{p} \alpha_i \ln \mu_{t+1-i|t-i} + \sum_{j=1}^{q} \beta_j \left(y_{t+1-j} \mu^{-1}_{t+1-j|t-j} - 1 \right)$$

$$\Rightarrow \mu_{t+1|t} = \prod_{i=1}^{p} \mu_{t+1-i|t-i}^{\alpha_i} \cdot e^{\omega + \sum_{j=1}^{q} \beta_j \left(\frac{y_{t+1-j}}{\mu_{t+1-j|t-j}} - 1 \right)} \quad //$$

Item c) Equação da média com CNO

$$
\begin{aligned}
p(y_t|y_{t-1}) &\sim NB(\mu_{t|t-1}, r) \\
E[y_t|y_{t-1}] &= \mu_{t|t-1} \\
h(\mu_{t+1|t}) &= \tilde{\mu}_{t+1|t} \\
\tilde{\mu}_{t+1|t} &= m_{t+1|t} + \gamma_{t+1|t}
\end{aligned}
$$

i) Sazonalidade por dummy Harrison & Stevens

1c) i) $p(y_t|y_{t-1}) \sim NB(\mu_{t|t-1}, r)$, $\mu_t > 0$ e $r > 0$

$E(y_t|y_{t-1}) = \mu_{t|t-1}$

$\tilde{\mu}_{t|t-1} = \ln \mu_{t|t-1} \Rightarrow \mu_t = e^{\tilde{\mu}_t}$, $\tilde{\mu}_t \in \mathbb{R}$

$$
\begin{cases}
\tilde{\mu}_{t+1|t} = m_{t+1|t} + \underset{\sim}{D}_{t+1} \underset{\sim}{\gamma}_{t+1|t} \\
m_{t+1|t} = \omega + \phi\, m_{t|t-1} + \kappa^m \tilde{s}_t \quad \text{, onde} \\
\underset{\sim}{\gamma}_{t+1|t} = \underset{\sim}{\gamma}_{t|t-1} + \underset{\sim}{K}{}_t^{\gamma}\, \tilde{s}_t
\end{cases}
$$

$$
\begin{cases}
\underset{\sim}{\gamma}_{t+1|t} = (\gamma_{1,t+1|t}, \gamma_{2,t+1|t}, \dots, \gamma_{12,t+1|t})' \\
\underset{\sim}{D}_t = (d_{1t}, d_{2t}, \dots, d_{12t})', \quad d_{jt} = 1 \text{ se } j = t \text{ e } d_{jt} = 0 \text{ c.c.} \\
\underset{\sim}{\tilde{K}}{}_t^{\gamma} = (\kappa_{1t}, \kappa_{2t}, \dots, \kappa_{12t}), \quad \kappa_{jt} = \kappa^{\gamma} \text{ se } j = t \text{ e } \kappa_{jt} = -\frac{\kappa^{\gamma}}{11} \text{ c.c.}
\end{cases}
$$

$\tilde{s}_t = y_t e^{-\tilde{\mu}_{t|t-1}} - 1$

ii) Sazonalidade por trigonométricos

A tendência $m_{t+1|t}$ segue um AR(1) com drift, enquanto a sazonalidade é dada pela soma de funções trigonométricas. É possível escrever as equações dessas componentes não observáveis na forma matricial abaixo.

$$
\begin{cases}
\tilde{\mu}_{t+1|t} = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} m_{t+1|t} \\ \gamma_{1,t+1|t} \\ \gamma^*_{1,t+1|t} \\ \gamma_{2,t+1|t} \\ \gamma^*_{2,t+1|t} \\ \gamma_{3,t+1|t} \\ \gamma^*_{3,t+1|t} \\ \gamma_{4,t+1|t} \\ \gamma^*_{4,t+1|t} \\ \gamma_{5,t+1|t} \\ \gamma^*_{5,t+1|t} \\ \gamma_{6,t+1|t} \end{pmatrix} \\
\begin{pmatrix} m_{t+1|t} \\ \gamma_{1,t+1|t} \\ \gamma^*_{1,t+1|t} \\ \gamma_{2,t+1|t} \\ \gamma^*_{2,t+1|t} \\ \gamma_{3,t+1|t} \\ \gamma^*_{3,t+1|t} \\ \gamma_{4,t+1|t} \\ \gamma^*_{4,t+1|t} \\ \gamma_{5,t+1|t} \\ \gamma^*_{5,t+1|t} \\ \gamma_{6,t+1|t} \end{pmatrix} = \begin{pmatrix} \omega \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix}
\alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & cos(\lambda_1) & sin(\lambda_1) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -sin(\lambda_1) & cos(\lambda_1) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & cos(\lambda_2) & sin(\lambda_2) & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -sin(\lambda_2) & cos(\lambda_2) & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & cos(\lambda_3) & sin(\lambda_3) & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -sin(\lambda_3) & cos(\lambda_3) & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & cos(\lambda_4) & sin(\lambda_4) & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -sin(\lambda_4) & cos(\lambda_4) & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & cos(\lambda_5) & sin(\lambda_5) & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -sin(\lambda_5) & cos(\lambda_5) & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1
\end{pmatrix} \cdot \begin{pmatrix} m_{t|t-1} \\ \gamma_{1,t|t-1} \\ \gamma^*_{1,t|t-1} \\ \gamma_{2,t|t-1} \\ \gamma^*_{2,t|t-1} \\ \gamma_{3,t|t-1} \\ \gamma^*_{3,t|t-1} \\ \gamma_{4,t|t-1} \\ \gamma^*_{4,t|t-1} \\ \gamma_{5,t|t-1} \\ \gamma^*_{5,t|t-1} \\ \gamma_{6,t|t-1} \end{pmatrix} + \begin{pmatrix} \kappa^m \\ \kappa^\gamma \\ \kappa^\gamma \\ \kappa^\gamma \\ \kappa^\gamma \\ \kappa^\gamma \\ \kappa^\gamma \\ \kappa^\gamma \\ \kappa^\gamma \\ \kappa^\gamma \\ \kappa^\gamma \\ \kappa^\gamma \end{pmatrix} \tilde{s}_t, \quad \lambda_j = \frac{2\pi j}{12}
\end{cases}
$$

$$
\begin{aligned}
\tilde{s}_t &= y_t e^{-\tilde{\mu}_{t|t-1}} - 1 \\
\Rightarrow \tilde{s}_t &= y_t e^{-m_{t|t-1} + \gamma_{t|t-1}} - 1
\end{aligned}
$$

Item d) Log da verossimilhança

1d) $\ell(\mu_t, \theta) = \sum_{t=1}^{T} \left(ln \binom{y_t + r - 1}{y_t} + r \, ln \, r + y_t \, ln \, \mu_t - (r + y_t) \, ln(r + \mu_t) \right)$

Como $r > 0$, usando link log

$\ell(\tilde{\mu}_t, \theta) = \sum_{t=1}^{T} \left(ln \binom{y_t + e^{\psi_1} - 1}{y_t} + \psi_1 e^{\psi_1} + y_t \tilde{\mu}_t - (r + y_t) \, ln(r + e^{\tilde{\mu}_t}) \right),$

$$
\begin{cases}
\theta = (\psi, \omega, \phi, \kappa^m, \kappa, m_1, \gamma_{j1}), \quad j = 1, \ldots, 12 \\
r = e^{\psi_1}, \; \psi_1 \in \mathbb{R} \\
\phi = 1 / (1 + e^{-\psi_2}), \; \psi_2 \in \mathbb{R} \Rightarrow 0 < \phi < 1
\end{cases}
$$

Questão 2

Item a)

2) $Y \sim f(y|\theta)$

a) $E(\nabla) = 0, \quad \nabla = \dfrac{\partial \ln f(y|\theta)}{\partial \theta}$

$$E(\nabla) = E\left(\frac{\partial \ln f(y|\theta)}{\partial \theta}\right)$$

$$= \int_{y \in \Omega} \frac{\partial \ln f(y|\theta)}{\partial \theta} f(y|\theta)\, dy \quad , \ \Omega \text{ suporte de } f(y|\theta)$$

$$= \int_{y \in \Omega} \frac{1}{\cancel{f(y|\theta)}} \frac{\partial f(y|\theta)}{\partial \theta} \cancel{f(y|\theta)}\, dy$$

$$= \frac{\partial}{\partial \theta} \int_{y \in \Omega} f(y|\theta)\, dy$$

$$= 0 \ , \text{ pois } \int_{y \in \Omega} f(y|\theta)\, dy = 1$$

Item b)

2b) $I = E(\nabla\nabla') = -E\left(\frac{\partial^2 \ln f(y|\theta)}{\partial\theta\,\partial\theta'}\right)$

$E(\nabla) = 0 \Rightarrow \int \frac{\partial \ln f(y|\theta)}{\partial\theta} f(y|\theta)\,dy = 0$

$\Rightarrow \frac{\partial}{\partial\theta} \int \frac{\partial \ln f(y|\theta)}{\partial\theta} f(y|\theta)\,dy = 0$

$\Rightarrow \int \left(\frac{\partial}{\partial\theta} \nabla f(y|\theta) + \nabla \frac{\partial f(y|\theta)}{\partial\theta} \right) dy = 0$

$\Rightarrow -\int \frac{\partial^2}{\partial\theta\,\partial\theta'} \ln f(y|\theta)\, f(y|\theta)\,dy = \int \nabla \underbrace{\frac{\partial \ln f(y|\theta)}{\partial\theta}}_{\nabla} f(y|\theta)\,dy$

$\Rightarrow -E\left(\frac{\partial^2 \ln f(y|\theta)}{\partial\theta\,\partial\theta'}\right) = E(\nabla\nabla') \quad //$

Questão 3

$$y_t = \mu_t + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma^2)$$

3)

$$\begin{cases} y_t = \mu_{t|t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0,\sigma^2) \\ \mu_{t+1|t} = \omega + \sum_{i=1}^{p} \alpha_i \mu_{t-i+1|t-i} + \sum_{j=1}^{q} \beta_j s_{t-j+1}, \quad s_t = (I_{t|t-1})^{-1} \nabla_t \end{cases}$$

• $p(y_t|y_{t-1}) \sim N(\mu_{t|t-1}, \sigma^2)$

$$\Rightarrow p(y_t|y_{t-1}) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{y_t-\mu_t}{\sigma}\right)^2}$$

$$\Rightarrow \ln p(y_t|y_{t-1}) = -\frac{1}{2} \ln 2\pi - \ln\sigma - \frac{1}{2} \frac{(y_t-\mu_t)^2}{\sigma^2}$$

$$\Rightarrow \nabla_t^{\mu} = \frac{y_t-\mu_t}{\sigma^2}, \quad \frac{\partial \nabla_t}{\partial\mu} = -\frac{1}{\sigma^2}$$

$$\Rightarrow I_{t|t-1} = -E\left(\frac{-1}{\sigma^2}\right) = \frac{1}{\sigma^2}$$

$$\Rightarrow s_t = \cancel{\sigma^2}\cdot\frac{(y_t-\mu_t)}{\cancel{\sigma^2}} = y_t - \mu_{t|t-1}$$

$$\Rightarrow \mu_{t+1|t} = \omega + \sum_{i=1}^{p} \alpha_i \mu_{t-i+1|t-i} + \sum_{j=1}^{q} \beta_j (y_{t-j+1} - \mu_{t-j+1|t-j})$$

• $\mu_{t|t-1} = y_t - \varepsilon_t \Rightarrow y_{t-j+1} - \mu_{t-j+1|t-j} = \varepsilon_{t-j+1}$

$$\Rightarrow \mu_{t+1|t} = \omega + \sum_{i=1}^{p} \alpha_i (y_{t-i+1} - \varepsilon_{t-i+1}) + \sum_{j=1}^{q} \beta_j \varepsilon_{t-j+1}$$

$$\Rightarrow y_{t+1} = \omega + \sum_{i=1}^{p} \alpha_i y_{t-i+1} + \sum_{j=1}^{q} \beta_j' \varepsilon_{t-j+1} + \varepsilon_{t+1},$$

$$\beta_j' = \begin{cases}\beta_j - \alpha_j, & j\le p \text{ e } j \le q \\ \beta_j, & p<j\le q \\ -\alpha_j, & q<j\le p\end{cases}$$

Questão 4

Item a) Parametrização e momentos da distribuição

4) $p(y_t | y_{t-1}) \sim \text{Weibull}(\mu, \kappa)$, $\kappa > 0$ e $\mu > 0$

• $E[y_t|y_{t-1}] = \lambda\Gamma(1+1/\kappa) = \mu \Rightarrow \lambda = \mu/\Gamma(1+1/\kappa)$

$\Rightarrow E[y_t|y_{t-1}] = \mu$

• $V[y_t|y_{t-1}] = \lambda^2 \left(\Gamma(1+2/\kappa) - (\Gamma(1+1/\kappa))^2\right)$

$= \dfrac{\mu^2}{\Gamma^2(1+1/\kappa)} \left(\Gamma(1+2/\kappa) - \Gamma^2(1+1/\kappa)\right)$

$= \mu^2 \left(\dfrac{\Gamma(1+2/\kappa)}{\Gamma^2(1+1/\kappa)} - 1\right)$

• $S(y_t|y_{t-1}) = \dfrac{\Gamma(1+3/\kappa)\lambda^3 - 3\mu\sigma^2 - \mu^3}{\sigma^3}$

$$= \dfrac{\dfrac{\Gamma(1+3/\kappa)\cancel{\mu^3}}{\Gamma^3(1+1/\kappa)} - 3\cancel{\mu^3}\left(\dfrac{\Gamma(1+2/\kappa)}{\Gamma^2(1+1/\kappa)} - 1\right) - \cancel{\mu^3}}{\cancel{\mu^3}\dfrac{\left(\Gamma(1+2/\kappa) - \Gamma(1+1/\kappa)\right)^{3/2}}{\Gamma^{3/2}(1+1/\kappa)}}$$

4a) continuação

$$\Rightarrow S(y_t|y_{t-1}) = \frac{\left(\frac{\Gamma(1+3/k)}{\Gamma^3(1+1/k)} - \frac{3\Gamma(1+2/k)}{\Gamma^2(1+1/k)} + 2\right)}{\left(\frac{\Gamma(1+2/k)}{\Gamma^2(1+1/k)} - 1\right)^{3/2}}$$

$$\bullet\ K(y_t|y_{t-1}) = \frac{\frac{\Gamma(1+4/k)}{\Gamma^4(1+1/k)} - 4\frac{\Gamma(1+3/k)}{\Gamma^3(1+1/k)} + 6\frac{\Gamma(1+2/k)}{\Gamma^2(1+1/k)} - 3}{\left(\frac{\Gamma(1+2/k)}{\Gamma^2(1+1/k)} - 1\right)^2}$$

Item b) Equação dos parâmetros

i) GAS

4b)

i) $p(y_t|y_{t-1}) \sim \text{Weibull}(\mu_{t|t-1}, K_{t|t-1})$

$E(y_t|y_{t-1}) = \mu_{t|t-1}$

$\tilde{\mu}_{t|t-1} = \ln(\mu_{t|t-1}) \Rightarrow$

$\tilde{K}_{t|t-1} = \ln(K_{t|t-1})$

$\tilde{\mu}_{t+1|t} = \omega^{\mu} + \alpha^{\mu}\tilde{\mu}_{t|t-1} + \beta^{\mu}\tilde{S}_t \quad , \quad \tilde{S}_t = \tilde{I}^{-1}_{t|t-1}\nabla_t$

$\tilde{K}_{t+1|t} = \omega^{K} + \alpha^{K}\tilde{K}_{t|t-1} + \beta^{K}\tilde{S}_t$

$$p(y_t|y_{t-1}) = \frac{K_{t|t-1}}{\lambda_{t|t-1}}\left(\frac{y_t}{\lambda_{t|t-1}}\right)^{K_{t|t-1}-1} e^{-\left(\frac{y_t}{\lambda_{t|t-1}}\right)^{K_{t|t-1}}} \quad , \quad y_t > 0$$

$$= \frac{K_t\Gamma_1}{\mu_t}\left(\frac{y_t\Gamma_1}{\mu_t}\right)^{K_t-1} e^{-\left(\frac{y_t\Gamma_1}{\mu_t}\right)^{K_t}}$$

$$\ln p(y_t|y_{t-1}) = \ln K_t\Gamma_1 + (K_t-1)\ln y_t\Gamma_1 - K_t\ln\mu_t - \left(\frac{y_t\Gamma_1}{\mu_t}\right)^{K_t}$$

4b) i) continuação

$$\frac{\partial \ell(y_t; F_{t-1}, \mu_t, \kappa_t)}{\partial \mu_t} = -\frac{\kappa_t}{\mu_t} + \frac{\kappa_t}{\mu_t}\left(\frac{y_t \Gamma_1}{\mu_t}\right)^{\kappa_t}$$

$$= \frac{\kappa_t}{\mu_t}\left(\left(\frac{y_t \Gamma_1}{\mu_t}\right)^{\kappa_t} - 1\right)$$

$$\frac{\partial \ell(y_t; F_{t-1}, \mu_t, \kappa_t)}{\partial \kappa_t} = \frac{\partial}{\partial \kappa_t}\left(\ln \kappa_t + \kappa_t \ln \Gamma_1 + (\kappa_t - 1)\ln y_t - \right.$$

$$\left. - \kappa_t \ln \mu_t - \left(\frac{y_t \Gamma_1}{\mu_t}\right)^{\kappa_t}\right)$$

$$\frac{\partial \kappa_t \ln \Gamma_1}{\partial \kappa_t} = \frac{\partial}{\partial \kappa_t} \kappa_t \ln \Gamma\left(1 + \frac{1}{\kappa_t}\right)$$

$$= \ln \Gamma_1 + \kappa_t \frac{\partial}{\partial \kappa_t} \ln \Gamma\left(1 + \frac{1}{\kappa_t}\right)$$

$$= \ln \Gamma_1 + \frac{\kappa_t}{\Gamma_1} \frac{\partial}{\partial \kappa_t} \Gamma\left(1 + \frac{1}{\kappa_t}\right)$$

$$= \ln \Gamma_1 + \frac{\cancel{\kappa_t}}{\cancel{\Gamma_1}} \cancel{\Gamma_1} \Psi\left(1 + \frac{1}{\kappa_t}\right)\left(-\frac{1}{\kappa_t^2}\right)$$

$$= \ln \Gamma_1 - \frac{\Psi_1}{\kappa_t}$$

$$\frac{\partial}{\partial \kappa_t}\left(\frac{y_t \Gamma_1}{\mu_t}\right)^{\kappa_t} = \frac{\partial}{\partial \kappa_t} e^{\kappa_t \ln \frac{y_t \Gamma_1}{\mu_t}} = \left(\frac{y_t \Gamma_1}{\mu_t}\right)^{\kappa_t}\left(\frac{\partial \kappa_t \ln \Gamma_1}{\partial \kappa_t} + \ln \frac{y_t}{\mu_t}\right)$$

$$\Rightarrow \frac{\partial \ell(y_t; F_{t-1}, \mu_t, \kappa_t)}{\partial \kappa_t} = \frac{1}{\kappa_t} + \ln \Gamma_1 - \frac{\Psi_1}{\kappa_t} + \ln y_t - \ln \mu_t - \left(\frac{y_t \Gamma_1}{\mu_t}\right)^{\kappa_t}\left(\ln \Gamma_1 - \frac{\Psi_1}{\kappa_t} + \right.$$

$$\left. + \ln y_t/\mu_t\right)$$

$$= \frac{1}{\kappa_t} + \ln \frac{y_t \Gamma_1}{\mu_t}\left(1 - \left(\frac{y_t \Gamma_1}{\mu_t}\right)^{\kappa_t}\right) + \frac{\Psi_1}{\kappa_t}\left(\left(\frac{y_t \Gamma_1}{\mu_t}\right)^{\kappa_t} - 1\right) \quad //$$

4b) i) continuação

$$\tilde{\nabla}_t = (\dot{h})^{-1} \nabla_t$$

$$\dot{h}(\mu_t) = \frac{\partial}{\partial \mu_t} \ln \mu_t = \frac{1}{\mu_t} \quad e \quad \dot{h}(\kappa_t) = \frac{\partial}{\partial \kappa_t} \ln \kappa_t = \frac{1}{\kappa_t}$$

$$\Rightarrow \begin{cases} \tilde{\nabla}_t^{\mu} = \mu_t \nabla_t^{\mu} \\ \tilde{\nabla}_t^{\kappa} = \kappa_t \nabla_t^{\kappa} \end{cases}$$

$$\Rightarrow \begin{cases} \tilde{\nabla}_t^{\mu} = \kappa_t \left(\left(\frac{y_t \Gamma_1}{\mu_t} \right)^{\kappa_t} - 1 \right), \quad \Gamma_1 = \Gamma\left(1 + \frac{1}{\kappa_t}\right) \\ \tilde{\nabla}_t^{\kappa} = 1 + \kappa_t \ln\left(\frac{y_t \Gamma_1}{\mu_t}\right) \left(1 - \left(\frac{y_t \Gamma_1}{\mu_t} \right)^{\kappa_t} \right) + \Psi_1 \left(\left(\frac{y_t \Gamma_1}{\mu_t} \right)^{\kappa_t} - 1 \right), \quad \Psi_1 = \overset{\text{digama}}{\Psi}\left(1 + \frac{1}{\kappa_t}\right) \end{cases}$$

$$\Rightarrow \begin{cases} \tilde{\nabla}_t^{\mu} = e^{\tilde{\kappa}_t} \left(\left(\frac{y_t \Gamma_1}{e^{\tilde{\mu}_t}} \right)^{e^{\tilde{\kappa}_t}} - 1 \right) \\ \tilde{\nabla}_t^{\kappa} = 1 + e^{\tilde{\kappa}_t} (\ln y_t \Gamma_1 - \tilde{\mu}_t) \left(1 - \left(\frac{y_t \Gamma_1}{e^{\tilde{\mu}_t}} \right)^{e^{\tilde{\kappa}_t}} \right) + \Psi_1 \left(\left(\frac{y_t \Gamma_1}{e^{\tilde{\mu}_t}} \right)^{e^{\tilde{\kappa}_t}} - 1 \right) \end{cases}$$

ii) CNO

4b) ii)

$$\begin{cases} p(y_t|y_{t-1}) \sim \text{Weibull}(\mu_{t|t-1}, \kappa), & \mu_t > 0 \text{ e } \kappa > 0 \\ E(y_t|y_{t-1}) = \mu_{t|t-1} \end{cases}$$

$$\tilde{\mu}_{t|t-1} = \ln \mu_{t|t-1} \Rightarrow \mu_t = e^{\tilde{\mu}_t}, \quad \tilde{\mu}_t \in \mathbb{R}$$

$$\begin{cases} \tilde{\mu}_{t+1|t} = m_{t+1|t} + \gamma_{t+1|t} \\ m_{t+1|t} = \omega + \phi\, m_{t|t-1} + \kappa^m s_t \\ \gamma_{t+1|t} = \sum_{j=1}^{6} \gamma_{j,t+1|t} \\ \begin{pmatrix} \gamma_{j,t+1|t} \\ \gamma^*_{j,t+1|t} \end{pmatrix} = \begin{pmatrix} \cos\lambda_j & \sin\lambda_j \\ -\sin\lambda_j & \cos\lambda_j \end{pmatrix} \begin{pmatrix} \gamma_{j,t|t-1} \\ \gamma^*_{j,t|t-1} \end{pmatrix} + \begin{pmatrix} \kappa^\gamma \\ \kappa^\gamma \end{pmatrix} s_t, \quad \lambda_j = \frac{2\pi j}{12} \end{cases}$$

Item c) Log da verossimilhança

4c)

$$\ell(\tilde{\mu}_t, \Theta) = \sum_{t=1}^{T} \ln p(y_t | y_{t-1}; \tilde{\mu}_t, \kappa)$$

$$= \sum_{t=1}^{T} \left(\ln\kappa - \kappa \ln\mu_t + \kappa \ln y_t \Gamma_1 - \left(\frac{y_t \Gamma_1}{\mu_t}\right)^{\kappa} \right), \quad \Gamma_1 = \Gamma\left(1 + \frac{1}{\kappa}\right)$$

Como $\kappa > 0$, usando link log

$$\kappa = e^{\psi}, \quad \psi \in \mathbb{R}$$

$$\Rightarrow \ell(\tilde{\mu}_t, \Theta) = \sum_{t=1}^{T} \left(\psi - e^{\psi} \tilde{\mu}_t + e^{\psi} \ln y_t \Gamma_1 - \left(\frac{y_t \Gamma_1}{e^{\tilde{\mu}_t}}\right)^{e^{\psi}} \right), \quad \Gamma_1 = \Gamma(1 + e^{-\psi})$$

com $\Theta = (\kappa, \psi, \omega, \phi, \kappa^m, \kappa^\gamma, m_1, \gamma_{i,1}, \gamma^*_{j,1})$, $i = 1, \dots, 6$
$j = 1, \dots, 5$

Item d) Estimação do modelo [TODO]